

Agenda

- ① Arithmetic Instruction
- ② Classification of operators
- ③ Unary operators
- ④ Arithmetic operators
- ⑤ Bitwise operators
- ⑥ Relational operators
- ⑦ Logical operators
- ⑧ Assignment operators

Arithmetic Instruction

An instruction which is used to manipulate data using operators, is known as Arithmetic Instruction.

operator

operands/data

$$3 + 4$$

- ① Unary
- ② Binary
- ③ Ternary

$$3 + 4 * 5$$

$$= 23$$

$$35$$

Classification of Operators

- ① Unary operators $+, -, ++, --, sizeof()$
- ② Arithmetic operators $*, /, \%, +, -$
- ③ Bitwise Operators $\&, |, \wedge, \sim, \gg, \ll$
- ④ Relational Operators $<, >, <=, >=, ==, !=$
- ⑤ Logical Operators $!, \&\&, ||$
- ⑥ Conditional Operator $?:$
- ⑦ Assignment Operators $=, +=, -=, *=, /=, \%=$

Unary Operators

+, -, ++, --

+5

-7

Increment operator

++

int x = 5;

x
~~7~~~~5~~~~6~~

printf("%d", x); 5

x++; $\leftarrow x = x + 1$ post increment

printf("%d", x); 6

++x; $\leftarrow x = x + 1$ pre increment

printf("%d", x); 7

x-- : post decrement

--x : pre decrement

x = x - 1

3++; error

Find Output of the following program?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int x = 5, y;
```

```
    y = x++;
```

```
    printf("%d %d", x, y);
```

```
}
```



6 5

Find Output of the following program?

```
#include <stdio.h>
```

```
int main()
```

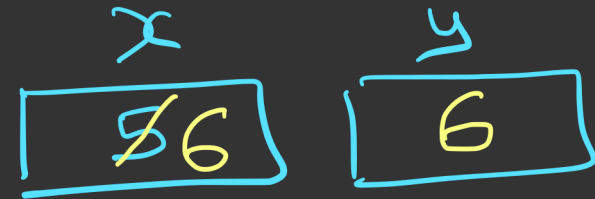
```
{
```

```
    int x = 5, y;
```

```
    y = ++x;
```

```
    printf("%d %d", x, y);
```

```
}
```



6 6

Unary Operators

sizeof()



- ① Data type
- ② variable
- ③ Constant

- Integer constant is of type int
- Real Constant is of type double
- Character constant is of type
int

Type modifiers

short

long

short int x;

short x;

char

1 byte

short

2 bytes

int

4 bytes

long

8 bytes

Arithmetic Operators

Associativity Rule: L to R

* / %

+ -

$$3+4 \quad 7$$

$$3-4 \quad -1$$

$$3*4 \quad 12$$

$$3/4 \quad 0$$

$$11/4 \quad 2$$

$$15\%6 \quad 3$$

$$21\%7 \quad 0$$

$$426\%10 \quad 6$$

$$3\%4 \quad 3$$

$$a * b / c$$

$$a / b * c$$

$$a + b * c$$

$$3/4 \rightarrow 0.75 \quad \leftarrow \text{maths}$$

Integer / Integer $\rightarrow$ integer

$$\begin{array}{r} 4 \overline{) 3} (0 \\ \underline{0} \\ 3 \end{array}$$

3.5 % 2 Error

$$3.0 / 4 \rightarrow 0.75$$

$$3 / 4.0 \rightarrow 0.75$$

$$3.0 / 4.0 \rightarrow 0.75$$

Bitwise Operators

$\&$, $|$, $\wedge$, $\sim$, $\gg$, $\ll$

```
int x;  
x = 35 & 52;
```

$$0 \& 0 \rightarrow 0$$

$$0 \& 1 \rightarrow 0$$

$$1 \& 0 \rightarrow 0$$

$$1 \& 1 \rightarrow 1$$

$$0 \wedge 0 \rightarrow 0$$

$$0 \wedge 1 \rightarrow 1$$

$$1 \wedge 0 \rightarrow 1$$

$$1 \wedge 1 \rightarrow 0$$

$$35 = 00100011$$

$$52 = 00110100$$

$$32 = 00100000$$

$$x = 44 | 68;$$

$$44 = 00101100$$

$$68 = 01000100$$

$$108 = 01101100$$

$$0 | 0 \rightarrow 0$$

$$0 | 1 \rightarrow 1$$

$$1 | 0 \rightarrow 1$$

$$1 | 1 \rightarrow 1$$

$$\sim 1 \rightarrow 0$$

$$\sim 0 \rightarrow 1$$

160

$$x = 53 \gg \underline{2};$$

00 00000000 00000000 00000000 00110101



13

$$x = 12 \ll 3;$$

~~00000000~~ 00000000 00000000 0001100000



1100000

$$64 + 32 = 96$$

Relational Operators (L+R)

$<$, $>$, $<=$, $>=$,

$==$, $!=$

Always give result
as 0 or 1

True $\rightarrow$ 1
False $\rightarrow$ 0

$3+4+5$
 $7+5$
Intersection: 4

$$0 \quad x = 3 > 4;$$

$$1 \quad x = 5 <= 5;$$

$$0 \quad x = 4 != 4;$$

$$0 \quad x = \underbrace{10 > 8}_{1} > 6;$$

$$1 \quad x = 3 != 4;$$

$$x = 2 == '2'$$

$$0 \quad x = 2 == 50$$

Logical Operators

! NOT (unary)

&& AND

|| OR

Every non-zero
value is True
Zero is False

! True $\rightarrow$ False (0)
! False $\rightarrow$ True (1)

exp1 && exp2

T	&& T	$\rightarrow$ T
T	&& F	$\rightarrow$ F
F	&& X	$\rightarrow$ F

exp1 || exp2

F	F	$\rightarrow$ F
F	T	$\rightarrow$ T
T	X	$\rightarrow$ T

x = ! 5 > -2;

! 5

0 > -2

!(5 > -2)

!!

0

$x > 0 \ \&\& \ y > 0$

T

T

F

F

T

F

T

F

$\rightarrow$ T

$\rightarrow$ F

$\rightarrow$ F

$\rightarrow$ F

Assignment Operators R to L

=

$x = 4 ;$ ✓

$4 = x ;$ ✗

$x = y = 4 ;$

variable =

$\text{int } x = 3 ;$

$x = x + 5 ;$

x

3

↑
Container

↑
Content

$= 3 + 5$

Compound Assignment Operators

$+=$, $-=$, $*=$, $/=$, $\%=$

`int x = 5;`

`x += 3;` $\rightarrow x = x + 3$

`x -= 4;` $\rightarrow x = x - 4;$

`x *= 7;` $\rightarrow x = x * 7$

`x /= 5;` $\rightarrow x = x / 5$

`x %= 2;` $\rightarrow x = x \% 2$